



CPSC 375: Introduction to Data Science and Big Data

CPSC 375-04

Fall 2026

Catalog Descriptions and Objectives

Learn techniques for data preparation, exploratory analysis, statistical modeling, machine learning, and visualization. Explore methods for analyzing different data types, such as natural language and time series, from emerging applications, including Internet-of-Things. Study big data platforms. Projects with real-world data.

Prerequisites

CPSC 131, MATH 338.

Undergraduate Course not available for Graduate Credit

G.E. Requirements

This class does not meet any CSU General Education requirements.

Learning Goals

1. Understand the basic concepts and methods of data science and big data analytics
2. Apply data science techniques to solve a well-defined data problem of moderate complexity
3. Recognize the relative strengths and weaknesses of common data analytics methods and determine the best choice for a given problem
4. Gain experience in applying data science methods to problems of personal interest
5. Function effectively in teams to accomplish a common goal via projects

Instructor

Professor Kanika Sood (You can **call me Dr. Sood or Dr. K !**)

Email: kasood@fullerton.edu (Email is the best way to reach me)

Office: CS 530 (In-person and on Zoom)

<https://fullerton.zoom.us/j/85718270183>

Tuesdays and Thursdays: 2:30-3:30 PM (In person and Zoom)

Fridays: 12:00PM-1:00PM (In person only)

Zoom details: Also Available on Canvas

Zoom Office Hours Details:

<https://fullerton.zoom.us/j/88231396345>

Passcode: 130389

If these times do not work for you, please send me an email and schedule an appointment time outside of office hours. During final exam week, office hours are by appointment only.

Meeting Information

Room: EC 034

Time: Friday 2:00PM - 4:45PM

Please note: Up to 20 % of the classes will be online via Zoom or a prerecording will be made available. Zoom details will be shared before the online sessions.

Exam Schedule

Midterm: Week 09

Final: 12/18/2026, Friday

Final Exam Time: 3:00PM - 4:50PM

Exam: Lecture Room

Response Time

The instructor usually responds to all emails within two working days, but occasionally may take longer than that. Responses to Canvas messages may have longer response time.

Textbooks

Required (all online and free)

[B1] [Python for Data Science](#), Arthur Turrell and [Contributors to Python4DS](#)

[B2] [Learning Data Science](#), Sam Lau, Joey Gonzalez, and Deb Nolan

- [B3] [Mining of Massive Datasets](#), 3rd edition, Jure Leskovec, Anand Rajaraman, and Jeffrey D. Ullman
- [B4] [Introduction to pyspark](#), First Edition, Pedro Duarte Faria

Optional

- [R for Data Science](#), Hadley Wickham and Garrett Grolemund
- [A Whirlwind Tour of Python](#), Jake VanderPlas
- [Python for Data Analysis, 3E](#), Wes McKinney
- [Python Data Science Handbook](#), Jake VanderPlas

Many popular technical books may be read online through the campus's subscription to [OReilly \(previously Safari\) Tech Books](#). From outside of the campus network, the campus library's WWW proxy will grant you access.

No Use of AI tools permitted

This course assumes that work submitted for a grade by students – all process work, drafts, brainstorming artifacts, final works – will be generated by the students themselves, working individually or in groups as directed by class assignment instructions. This policy indicates the following constitute violations of academic honesty: a student has another person/entity do the work of any substantive portion of a graded assignment for them, which includes purchasing work from a company, hiring a person or company to complete an assignment or exam, and/or using generative AI tools (such as ChatGPT).

In this course, every element of class assignments must be fully prepared by the student. The use of generative AI tools for any part of your work will be treated as plagiarism. If you have questions, please contact me.

Course Outline (Subject to change)

Week	Material	Textbook Sections
Week 1	What is data science? Introduction to Python	B1: Introduction B1: Ch. 1 B1: Ch. 10
Week 2	Visualization	B1: Ch. 2 B1: Ch 11-12
Week 3	Exploratory data analysis	B1: Ch. 3-4 B1: Ch. 7 B1: Ch. 9 B1: Ch. 13

Week 4	Data wrangling	B1: Ch. 4
Week 5	Tidy data	B1: Ch. 22 B1: Ch. 6 B1: Ch. 21
Week 6	Linear modeling	B2: Ch. 4 B2: Ch. 15
Week 7	Nearest neighbors classification Distance measures	B3: Ch. 12.4
Week 8	K-means clustering Evaluation metrics Midterm Review	B3: Ch. 7.3.1-7.3.3 B2: Ch. 19.5 B2: Ch. 16.2-16.3
Week 9	Evaluation metrics (contd.) Midterm exam	<i>N/A</i>
Week 10	Introduction to IoT; Time-series data (Intro.)	Lecture slides
Week 11	Text data	B2: Ch. 13
Week 12	Text data (contd.)	B2: Ch. 21
Week 13	Data storage	B3: Ch. 2.1
Week 14	Fall Recess (November 23 -29)	
Week 15	Big data platforms (MapReduce)	B3: Ch. 2.2
Week 15	Big data platforms (Apache Spark)	B3: Ch. 2.4.2-2.4.3 B4
Finals	Final exam (covers material after midterm)	<i>N/A</i>

Development Tool Resources

The course introduces students to data analysis, statistics, and plotting using the Python language. Prior knowledge of Python is not required. The recommended Python development environments are [Visual Studio Code](#) with [Anaconda](#) and [Google Colaboratory](#).

For the Big Data portion of the course, students will be expected to install Apache Spark on a Linux platform. For this, it is recommended to use the Computer Science Department's official Linux

development environment, Tuffix. Tuffix is Tuffy the Titan's Linux distribution. You can set up Tuffix as a native install on a dedicated computer, or as a virtual machine (VM), by following the [Tuffix Installation Instructions](#). You can use your own computer, or borrow a computer from CSUF for free through the [Long-Term Laptop Checkout](#) process.

Another option is CSUF's [Virtual Computing Lab \(VCL\)](#). This allows you to connect remotely to a Tuffix-like environment. Keep in mind a VCL session only lasts a limited time (up to 4 hours) and *is erased at the end of the session*, so you *must save your work elsewhere before your session ends* (e.g. git push all your work before the session ends).

Students using Tuffix should join the [CSUF TUFFIX](#) slack workspace at <https://csuf-tuffix.slack.com>. Please use the #general channel to ask about troubleshooting, installing, and using Tuffix.

Students are also expected to

1. Have basic computer competency which includes:
 - a. the ability to use a personal computer to locate, create, move, copy, delete, name, rename, and save files and folders on hard drives, secondary storage devices such as USB drives, and cloud such as Google Drive (Titan Apps);
 - b. the ability to use a word processing program to create, edit, format, store, retrieve, and print documents;
 - c. the ability to use their CSUF email accounts to receive, create, edit, print, save, and send an e-mail message with and without an attached file; and
 - d. the ability to use an Internet browser to search and access web sites in the World Wide Web.
2. Have ongoing reliable access to a computer with Internet connectivity for regular course assignments
3. Access their CSUF student email account
4. Use Internet search and retrieval skills to complete assignments
5. Apply their educational technology skills to complete expected competencies
6. Utilize other software applications as course requirements dictate

Important Dates

CSUF's Academic Calendar is posted online at <http://apps.fullerton.edu/AcademicCalendar/>. The Academic Calendar contains all the campus closures and holidays you should be aware of. CSUF's Admissions Calendar is posted online at <http://www.fullerton.edu/admissions>. The Admissions Calendar contains all the major dates with respect to adding, dropping, and withdrawing from your classes.

Grading

The final grade will be based on the following breakdown:

Category	% of Final Grade
Homework	25 %

Course participation	5 %
Projects	20 %
Midterm	25 %
Final	25 %

This class calls on you to work independently as well as collaboratively. **Please note there are assignments to assess your abilities both as an individual and as a member of a group.** If you have a question about what collaboration, group projects, or individual work should look like, please do not hesitate to come to see me. Plus and minus grading are not used when determining final grades.

Final grades are computed by first finding the average score in each category described in the table below on the right. All scores are normalized to a scale of 0 to 100 before being averaged. The average score for each category is then used to compute the weighted average according to the weights in the second table below.

In this course the plus/minus system will be used. The grade breakdown is as follows:

98 – 100% = A+
 93 - 97% = A (outstanding performance)
 90 – 92% = A-
 87 – 89% = B+
 83 – 86% = B (good performance)
 80 – 82% = B-
 77 – 79% = C+
 73 – 76% = C (acceptable performance)
 70 – 72% = C-
 67- 69% = D+
 63 – 66% = D (poor performance)
 60 – 62% = D-
 0 – 59% = F

The instructor reserves the right to adjust the grades up according to the performance of the whole class. However, there will not be any exceptions for any individual students. Please check the class Canvas site for class announcements, homework assignments, and your current status in the class.

Assignments

Programming and written assignments will be discussed in class and posted to the course website in advance of their due dates.

All programming assignments must be written in the programming language(s) specified as part of the assignment. Coding style must conform to professional norms. At a minimum, code must be commented,

have descriptive names for identifiers, and contain a comment with pertinent information such as the student's name, email address, and assignment name.

Homework

Students should form groups of 2-3 people. Each group will be responsible for completing each assignment without collaborating with other groups in the class. The outcome of each homework assignment will be a written report and/or source code with the solutions to the assigned problems (one per group) and must be submitted electronically on Canvas.

Projects

There will be one projects on Data Science/Big Data and will involve designing, implementing, and analyzing a substantial program in Python.

Examinations

The midterm and final are **not** cumulative. All exams are to be completed individually. Please be on time for the exams. Additional aids (e.g., calculators) or support materials may be used as described by the instructor prior to each exam. Make-up exams will be considered only in the most extraordinary of circumstances, such as serious illness or accident (written documentation will be required).

Alternative Procedure for Submitting Work

In case of technical difficulties with Canvas, the instructor will communicate with students directly through CSUF email, and assignments can be sent through email. In the case email doesn't work, students should call the Computer Science department coordinator at 657-278-3700 for further direction.

Make Up Policy

The following kinds of projects cannot be evaluated, and will be assigned a zero score:

- Late submissions.
- Email submissions.
- Source code that cannot be executed successfully.
- Input/output that is falsified or does not match the submitted source code.
- Submissions that are plagiarized or otherwise violate the collaboration guidelines.

Extra Credit

There are no opportunities for extra credit.

Collaboration

Collaboration is *not* allowed on any **exam**. You may work on **homework** and **projects** in groups of **up to three**. When you work in a group, make one submission with all group members' names. You may work freely with your fellow group members, but must limit the input you get from sources *outside* your group:

- You may help each other understand the assignment and brainstorm general solutions, but each group must develop and submit their own distinct work.
- You may give each other technical support, for instance troubleshooting installing Apache Spark.
- You must separate to develop your own detailed solution to the problem, and type in your own source code and project report.
- Given these requirements, any submissions with identical excerpts, or excerpts that are identical up to superficial rearrangements, will be considered highly suspect of plagiarism.

Administrative drops

Any student who misses the first class meeting may be dropped from the class unless they contact the instructor or the Computer Science department within 24 hours.

Attendance Policy

Attending class is mandatory. Missing class is not allowed unless it is excused by the instructor. Missing class as part of a documented accommodation is guaranteed to be excused. The ADA accommodated student must make a reasonable effort to coordinate any absences with the instructor. Please keep in mind that this course will often be run like a seminar; that means both your attendance and your participation in-class activities are required for successful completion of the course. Plan to arrive on time and stay until the end of class. On days we have peer review, plan to bring copies of the draft and copies for your peer group. If you need to arrive late or leave early, please try to minimize disruptions.

Participation

In the context of this course, participation is defined as the following:

- Arriving to class prepared and on time.
- Taking notes.
- Actively listening to the lecture and asking questions when appropriate.
- Annotating code listings and handouts.
- Bringing any required materials to class.

- Participating actively in class discussions.
- When needed/desired, seek assistance to complete assignments.
- Barring an emergency, not leaving the class session early unless the instructor consents.
- Not distracting oneself or others with smartphones, games, online diversions, etc.
- Respecting and treating the instructor and the student's peers civilly.

Note: To score full participation points you need to actively participate each week in almost each session.

Announcements/Communication

Announcements will be made via email to each student's @fullerton.edu email addresses, as well as via Canvas (course management system). You have a CSUF-supplied email account, and that is the only way I have of reaching you outside class. Check that account daily for important class announcements and individual emails. I try to respond to all emails within one working day, but occasionally it may take longer than that. Plan accordingly, especially around deadlines.

Academic Dishonesty

Students are encouraged to assist one another and discuss the course materials with their peers. It is your responsibility to be aware of and follow the spirit of CSU Fullerton's academic honesty policy which can be found at

http://www.fullerton.edu/senate/publications_policies_resolutions/ups/UPS%20300/UPS%20300.021.pdf.

Academic dishonesty will not be tolerated. The University Catalog and the Class Schedule provide a detailed description of Academic Dishonesty under *University Regulations*.

By submitting work for evaluation, you acknowledge that you have adhered to the spirit of the university's academic honesty policy and that your submission is an original work by you unless otherwise directed to work in groups. Failure to follow the spirit of the academic honesty policy will result in a severely negative evaluation of the work in question and may result in involving the Department Chair and the Judicial Affairs office to seek a disciplinary remedy.

Plagiarism is defined as the act of taking the specific substance of another and offering it as one's own without giving credit to the source (e.g., copying another person's program).

When you use sources, you must acknowledge the original author or source following standard scholarly practice. You are not allowed to use any material from any website that provides solutions to the assignments given in class.

Failure to follow the spirit of the academic honesty policy will result in a severely negative evaluation of your work in question. Each offense will be reported to the Department Chair and to the Dean of Students office, Student Conduct. A first offense will result in a zero score on the offending assignment. A subsequent offense will result in an F in the course.

Accessibility And Accommodations

Please inform the instructor during the first week of classes about any disability or special needs that you may have that may require specific arrangements related to attending class sessions, carrying out class assignments, or writing papers or examinations. According to California State University policy, students with disabilities must document their disabilities at the Disability Support Services (DSS) Office in order to be accommodated in their courses. Additional information can be found at <http://www.fullerton.edu/DSS/>, by calling 657-278-3112 or email dsservices@fullerton.edu.

Student Mental Health and Well-Being

We are still in an unusual time in our lives. I hope that you and your family and friends have remained healthy. I want to help you navigate through this period as I can. Listed below are resources for you to contact – do not hesitate if you need support. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, contact Counseling and Psychological Services (CAPS) 24/7 crisis phone services for students at (657) 278-3040 and <https://www.fullerton.edu/caps/>. Visit YOU @ Fullerton, a free and confidential wellness portal that connects students to academic, physical and mental health resources. Please call 911 for any and all emergencies.

Student Resources

Any student who wishes to discuss any concern may contact the assistant deans of the college. Assistant deans are student advocates who will help you navigate the university's policies and procedures and assist with resolving any conflict.

Assistant Dean for Student Affairs <https://www.fullerton.edu/ecs/dean/assistantdean.php> Shannen Allado
CS-206A (657) 278-4407 «shallado@fullerton.edu»

College International Advisor Sandra Boulanger
CS-206A (657) 278-3119 «sboulanger@fullerton.edu»

Emergency Procedures

For your own safety and the safety of others, each student is expected to read and understand the guidelines published at <http://prepare.fullerton.edu/campuspreparedness/>. Should an emergency occur, follow the instructions given to you by faculty, staff, and public safety officials. An emergency information recording is available by calling the Campus Operation and Emergency Closure line at 657-278-4444.

Instructional Continuity

Due to an event such as an epidemic or a natural disaster that disrupts normal campus operations, students must monitor the course Canvas site and their campus email address for any instructions and assignments that the instructor announces.

Recording & Transcription of Class Content

Recording class content is governed by UPS 330.230, http://www.fullerton.edu/senate/publications_policies_resolutions/ups/UPS%20300/UPS%20330.230.pdf. Each instructor must permit class content to be recorded or transcribed by students when mandated to do so by the Americans with Disabilities Act or by other federal or state laws. Any recording of class content is for private use and study and shall not be made publicly accessible without the written consent of the instructor and students in the class.

Course Rules & Classroom Management

Unless an agreement or accommodation is reached between the student and the instructor, these rules must be followed.

- Attendance at all regularly scheduled lectures and discussion sections is mandatory.
- Please put your cell phones on silent when coming to the class.
- Do not eat during lectures.
- If it makes noise, silence it.
- Computer use is not allowed in lectures except for taking notes.
- The student is responsible to be aware of any course announcements including changes to due dates and requirements.
- Third-party work (code, artwork, etc.) may not be used in student work without prior instructor consent. Failure to gain and document instructor consent will be construed as willful academic dishonesty.
- When a third party's work is incorporated into student work after gaining instructor consent, failure to wholly document the work's origin, copyright and license will be construed as willful academic dishonesty.

Library Support

The [Pollak Library](#) has many services to offer students.

Netiquette Requirements

Each student is expected to conduct themselves in a professional manner during the class - taking full advantage of the learning opportunities available. This includes completing all online discussions and assignments, adhering to proper netiquette, and so on. Netiquette refers to a set of behaviors that are appropriate for online activity - especially with email and threaded discussions. The core rules of netiquette can be found at the [Netiquette website](#). Please read through these netiquette rules to ensure that you are familiar with what will be the expected online behavior for this course.

Acknowledgment

Large portions of this syllabus draw from the syllabus authored by Kenytt Avery who coteaches this course.